Lesson 30: Interior-Point Methods

AI/ML Mathematics

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1 Introduction

Interior-point methods solve constrained convex optimization problems by converting hard inequality constraints into smooth barrier terms added to the objective. By systematically reducing the barrier weight, these methods trace a smooth curve—the *central path*—that converges to the constrained optimum. The resulting algorithms achieve polynomial-time complexity: $O(\sqrt{m}\log(1/\epsilon))$ Newton iterations for m inequality constraints and ϵ-accuracy.

Interior-point methods are particularly relevant for reinforcement learning because they efficiently solve the *linear programs* (LPs) and *semidefinite programs* (SDPs) that arise in exact MDP formulations and constrained policy optimization.

This lesson develops the barrier method from first principles. Section 2 motivates the approach. Section 3 defines the logarithmic barrier and establishes its convexity. Section 4 introduces the central path and proves the central path optimality bound. Section 5 presents the barrier method algorithm. Section 6 provides the complexity analysis. Section 7 briefly describes Newton's method for the centering step. Section 8 connects interior-point methods to the MDP LP formulation. Section 9 provides exercises.

The primary references are Boyd and Vandenberghe [2004] (Chapters 11) and Wright [1997].

2 Motivation

Consider the standard constrained convex optimization problem:

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} \; f_0(x) \\ &\text{subject to } f_i(x) \le 0, \quad i = 1,\dots,m, \\ &\qquad\qquad Ax = b, \end{aligned} \tag{1}$$

where $f_0, f_1, \dots, f_m : \mathbb{R}^n \to \mathbb{R}$ are convex and twice continuously differentiable, $A \in \mathbb{R}^{p\times n}$ with $\operatorname{rank}(A) = p$, and $b \in \mathbb{R}^p$.

First-order methods (gradient descent, proximal methods) handle *unconstrained* or simply constrained problems efficiently, but they struggle with general inequality constraints. Two standard approaches to handling constraints are:

(i) **Projection methods:** Project each iterate onto the feasible set. This requires solving a subproblem at each step and is efficient only when projection is cheap (e.g., onto boxes or balls).

(ii) **Barrier methods:** Encode constraints in the objective using a barrier function that goes to $+\infty$ at the constraint boundary. This converts the constrained problem into a sequence of *unconstrained* problems.

Interior-point methods follow approach (ii). Their distinguishing feature is polynomial worst-case complexity, unlike simplex-type methods (which can be exponential in the worst case) or first-order methods (which have slow convergence for ill-conditioned problems).

Constrained Optimization in RL

Constrained optimization is central to RL in several ways:

- **MDP as LP:** A finite MDP with $|S|$ states and $|A|$ actions can be solved exactly via a linear program with $|S|\cdot|A|$ variables and $|S|$ equality constraints plus $|S|\cdot|A|$ non-negativity constraints. Interior-point methods solve this in polynomial time.
- **Constrained MDPs:** Safe RL problems impose cost constraints $C(\pi) \le \alpha$ along-

side the return objective $J(\pi)$. The Lagrangian relaxation converts these into the form of (1).

- **Robust MDPs:** Minimax formulations with uncertainty sets lead to SDPs solvable by interior-point methods.

3 The Logarithmic Barrier

Definition 3.1 (Strictly feasible set)**.** The *strictly feasible set* of problem (1) (ignoring equality constraints for simplicity) is

$$\mathcal{F}^\circ = \{x \in \mathbb{R}^n : f_i(x) < 0,\ i = 1, \dots, m\}. \tag{2}$$

We assume $\mathcal{F}^\circ \neq \emptyset$ (Slater's condition).

Definition 3.2 (Logarithmic barrier)**.** The *logarithmic barrier function* for the inequality constraints in (1) is

$$\phi(x) = -\sum_{i=1}^{m} \log(-f_i(x)), \tag{3}$$

defined for $x \in \mathcal{F}^\circ$. We set $\phi(x) = +\infty$ for $x \notin \mathcal{F}^\circ$.

Remark 3.3 (Barrier behavior). The barrier $\phi(x)$ increases without bound as x approaches the boundary of the feasible set (i.e., as any $f_i(x) \to 0^-$). This "repels" the iterates from the constraint boundary, keeping them strictly feasible—hence the name *interior-point* method.

Proposition 3.4 (Convexity of the logarithmic barrier)**.** *If $f_1, \dots, f_m$ are convex, then the logarithmic barrier ϕ is convex on $\mathcal{F}^\circ$.*

Proof. It suffices to show that each term $-\log(-f_i(x))$ is convex on $\{x : f_i(x) < 0\}$.

Step 1: Composition rule. The function $\psi(u) = -\log(-u)$ is defined for $u < 0$. We compute:

$$\psi'(u) = \frac{-1}{-u} \cdot (-1) = \frac{1}{u} \quad \left(\text{wait: } \psi'(u) = \frac{d}{du}[-\log(-u)] = \frac{-1}{-u} \cdot (-1) = -\frac{1}{u}\right).$$

Let us be more careful. For $u < 0$: $\psi(u) = -\log(-u)$, so $\psi'(u) = -\frac{1}{-u} \cdot (-1) = -\frac{1}{u}$, and $\psi''(u) = \frac{1}{u^2} > 0$.

Since $\psi''(u) > 0$ for all $u < 0$, the function ψ is *strictly convex* on $(-\infty, 0)$.

Step 2: Convexity of composition. Moreover, ψ is *non-decreasing* on $(-\infty, 0)$: indeed, $\psi'(u) = -1/u > 0$ for $u < 0$. By the composition rule for convex functions [Boyd and Vandenberghe, 2004, Section 3.2.4]: if ψ is convex and non-decreasing, and f_i is convex, then $\psi \circ f_i$ is convex on $\{x : f_i(x) < 0\}$.

Step 3: Sum of convex functions. Each $-\log(-f_i(x))$ is convex on $\{x : f_i(x) < 0\}$. The sum $\phi(x) = \sum_{i=1}^m [-\log(-f_i(x))]$ is convex on $\mathcal{F}^\circ = \bigcap_{i=1}^m \{x : f_i(x) < 0\}$. □

Remark 3.5 (Gradient and Hessian of the barrier). The gradient and Hessian of ϕ are

$$\nabla\phi(x) = \sum_{i=1}^{m} \frac{1}{-f_i(x)} \nabla f_i(x), \tag{4}$$

$$\nabla^2\phi(x) = \sum_{i=1}^{m} \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^\top + \sum_{i=1}^{m} \frac{1}{-f_i(x)} \nabla^2 f_i(x). \tag{5}$$

The first term in the Hessian is always positive semidefinite (it is a sum of rank-one matrices scaled by positive coefficients). Near the boundary of the feasible set, where some $f_i(x) \to 0^-$, the Hessian grows without bound, reflecting the steep barrier.

Example 3.6 (Barrier for linear inequalities)**.** For the constraint set $\{x : a_i^\top x \le b_i\}_{i=1}^m$ (i.e., $f_i(x) = a_i^\top x - b_i$), the barrier is

$$\phi(x) = -\sum_{i=1}^m \log(b_i - a_i^\top x).$$

The gradient is $\nabla\phi(x) = \sum_i \frac{a_i}{b_i - a_i^\top x}$ and the Hessian is $\nabla^2\phi(x) = \sum_i \frac{a_i a_i^\top}{(b_i - a_i^\top x)^2} \succeq 0$.

4 The Central Path

Definition 4.1 (Barrier problem)**.** For a parameter $t > 0$, the *barrier problem* (or *penalized problem*) associated with (1) is

$$\min_x \; t\, f_0(x) + \phi(x) \quad \text{subject to } Ax = b, \tag{6}$$

where ϕ is the logarithmic barrier (3).

Remark 4.2 (Interpretation of t)*.* The parameter $t > 0$ controls the trade-off between optimizing the original objective f_0 and staying away from the constraint boundary (via ϕ). As $t \to \infty$, the objective $t\, f_0(x)$ dominates, and the minimizer of (6) approaches the constrained optimum of (1). As $t \to 0^+$, the barrier dominates, and the minimizer is the *analytic center* of the feasible set.

Definition 4.3 (Central path)**.** Assuming that for each $t > 0$, the barrier problem (6) has a unique minimizer $x^*(t)$, the curve $\{x^*(t) : t > 0\}$ is called the *central path*.

Proposition 4.4 (Central path characterization)**.** *Each point $x^*(t)$ on the central path satisfies the optimality conditions: there exists $\nu^*(t) \in \mathbb{R}^p$ such that*

$$t\,\nabla f_0(x^*(t)) + \nabla\phi(x^*(t)) + A^\top \nu^*(t) = 0, \qquad Ax^*(t) = b. \tag{7}$$

Proof. The barrier problem (6) is a convex optimization problem with equality constraints $Ax = b$. Its Lagrangian is $\mathcal{L}(x, \nu) = t f_0(x) + \phi(x) + \nu^\top(Ax - b)$. Since the problem is convex and (assuming constraint qualification) feasible, the KKT conditions are necessary and sufficient. The stationarity condition $\nabla_x \mathcal{L} = t\nabla f_0(x) + \nabla\phi(x) + A^\top \nu = 0$ combined with primal feasibility $Ax = b$ gives (7). □

Theorem 4.5 (Central path optimality bound)**.** *Let p^* denote the optimal value of* (1)*. Each point $x^*(t)$ on the central path satisfies*

$$\boxed{f_0(x^*(t)) - p^* \le \frac{m}{t},} \tag{8}$$

where m is the number of inequality constraints.

Proof. We construct a dual feasible point from the central path conditions and use weak duality.

Step 1: Define dual variables. From (7) and the formula for $\nabla\phi$ (4):

$$t\,\nabla f_0(x^*(t)) + \sum_{i=1}^m \frac{1}{-f_i(x^*(t))} \nabla f_i(x^*(t)) + A^\top \nu^*(t) = 0. \tag{9}$$

Define the dual variables

$$\lambda_i^*(t) = \frac{1}{-t\, f_i(x^*(t))} > 0, \quad i = 1, \ldots, m. \tag{10}$$

Note that $\lambda_i^*(t) > 0$ since $f_i(x^*(t)) < 0$ (the central path stays strictly feasible). Substituting into (9):

$$\nabla f_0(x^*(t)) + \sum_{i=1}^{m} \lambda_i^*(t) \nabla f_i(x^*(t)) + A^\top \left(\frac{\nu^*(t)}{t} \right) = 0. \tag{11}$$

This is precisely the stationarity condition for the Lagrangian $L(x, \lambda, \nu) = f_0(x) + \sum_i \lambda_i f_i(x) + \nu^\top (Ax - b)$ evaluated at $x = x^*(t)$, $\lambda = \lambda^*(t)$, $\nu = \nu^*(t)/t$.

Step 2: Evaluate the dual function. Since $x^*(t)$ minimizes $L(x, \lambda^*(t), \nu^*(t)/t)$ over x (by the stationarity condition), the dual function is

$$\begin{aligned} g(\lambda^*(t), \nu^*(t)/t) &= \inf_x L(x, \lambda^*(t), \nu^*(t)/t) \\ &= L(x^*(t), \lambda^*(t), \nu^*(t)/t) \\ &= f_0(x^*(t)) + \sum_{i=1}^{m} \lambda_i^*(t) f_i(x^*(t)) + \frac{\nu^*(t)^\top}{t} (Ax^*(t) - b). \end{aligned} \tag{12}$$

Since $Ax^*(t) = b$, the last term vanishes. For the middle sum:

$$\sum_{i=1}^{m} \lambda_i^*(t) f_i(x^*(t)) = \sum_{i=1}^{m} \frac{f_i(x^*(t))}{-t\, f_i(x^*(t))} = \sum_{i=1}^{m} \frac{-1}{t} = -\frac{m}{t}. \tag{13}$$

Therefore:

$$g(\lambda^*(t), \nu^*(t)/t) = f_0(x^*(t)) - \frac{m}{t}. \tag{14}$$

Step 3: Apply weak duality. By weak duality (the dual function provides a lower bound on the primal optimal value):

$$g(\lambda^*(t), \nu^*(t)/t) \le p^*. \tag{15}$$

Combining (14) and (15):

$$f_0(x^*(t)) - \frac{m}{t} \le p^*, \tag{16}$$

which rearranges to $f_0(x^*(t)) - p^* \le m/t$. □

Central Path Optimality Bound

Each point on the central path is within m/t of optimal: $f_0(x^*(t)) - p^* \le m/t$.

- The *duality gap* along the central path is exactly m/t (from (13), the complementary slackness residual is $-m/t$).
- To achieve ϵ-accuracy, we need $t \ge m/\epsilon$.
- The dual variables (10) approach complementary slackness as $t \to \infty$: $\lambda_i^*(t) f_i(x^*(t)) = -1/t \to 0$.

Remark 4.6 (Approximate complementary slackness). At the central path point $x^*(t)$, each product $\lambda_i^*(t) \cdot (-f_i(x^*(t))) = 1/t$ is the same for all i. This "uniform" near-complementarity is a distinctive feature of interior-point methods, in contrast to active-set methods where some constraints are exactly active and others are strictly inactive.

5 The Barrier Method

Definition 5.1 (Barrier method (path-following method))**.** The *barrier method* solves (1) by approximately following the central path. Given:

- A strictly feasible starting point $x^{(0)} \in \mathcal{F}^\circ$,
- An initial barrier parameter $t^{(0)} > 0$,
- A growth factor $\sigma > 1$,
- A tolerance $\epsilon > 0$,

the algorithm proceeds as follows:

$$\boxed{\begin{array}{l}\textbf{repeat } \ell = 0, 1, 2, \ldots \\ \quad 1.\ \textbf{Centering step:} \text{ Compute } x^{(\ell+1)} \approx x^*(t^{(\ell)}) \\ \quad\quad \text{by minimizing } t^{(\ell)} f_0(x) + \phi(x) \text{ subject to } Ax = b, \\ \quad\quad \text{starting from } x^{(\ell)}, \text{ using Newton's method.} \\ \quad 2.\ \textbf{Update: } t^{(\ell+1)} = \sigma \cdot t^{(\ell)}. \\ \quad 3.\ \textbf{Stop if } \ m/t^{(\ell)} \le \epsilon.\end{array}} \tag{17}$$

Remark 5.2 (Warm starting). The key to the efficiency of the barrier method is *warm starting*: the centering step at parameter $t^{(\ell+1)} = \sigma\, t^{(\ell)}$ starts from $x^{(\ell+1)}$, which is already close to $x^*(t^{(\ell)})$ (the previous central path point). If σ is not too large, then $x^*(t^{(\ell)})$ is close to $x^*(\sigma\, t^{(\ell)})$, so Newton's method converges in few iterations.

Proposition 5.3 (Number of outer iterations)**.** *The barrier method terminates after at most*

$$\left\lceil \frac{\log(m/(t^{(0)}\epsilon))}{\log \sigma} \right\rceil \tag{18}$$

outer iterations (centering steps).

Proof. The stopping criterion is $m/t^{(\ell)} \le \epsilon$, i.e., $t^{(\ell)} \ge m/\epsilon$. Since $t^{(\ell)} = \sigma^\ell\, t^{(0)}$, we need $\sigma^\ell \ge m/(t^{(0)}\epsilon)$, i.e., $\ell \ge \log(m/(t^{(0)}\epsilon))/\log \sigma$. □

6 Complexity Analysis

The overall complexity of the barrier method depends on two factors:

(i) The number of outer iterations (Proposition 5.3).

(ii) The number of Newton steps per centering computation.

The theory of *self-concordant functions* [Nesterov, 2004, Wright, 1997] provides the key to analyzing (ii).

Definition 6.1 (Self-concordant function)**.** A convex function $g : \mathbb{R} \to \mathbb{R}$ is *self-concordant* if $|g'''(x)| \le 2\, g''(x)^{3/2}$ for all $x \in \operatorname{dom} g$. More generally, $g : \mathbb{R}^n \to \mathbb{R}$ is self-concordant if for every $x \in \operatorname{dom} g$ and every direction $v \in \mathbb{R}^n$, the univariate function $t \mapsto g(x + tv)$ is self-concordant.

Proposition 6.2 (Logarithmic barrier is self-concordant)**.** *The logarithmic barrier $\phi(x) = -\sum_{i=1}^m \log(-f_i(x))$ is self-concordant when the constraint functions f_i are affine ($f_i(x) = a_i^\top x - b_i$).*

Proof. For a single linear constraint $f(x) = a^\top x - b$, the barrier term is $g(x) = -\log(b - a^\top x)$. Along any direction v: $\tilde{g}(t) = -\log(b - a^\top(x + tv)) = -\log(c - dt)$ where $c = b - a^\top x > 0$ and $d = a^\top v$. Then:

$$\tilde{g}'(t) = \frac{d}{c - dt}, \qquad \tilde{g}''(t) = \frac{d^2}{(c - dt)^2}, \qquad \tilde{g}'''(t) = \frac{2d^3}{(c - dt)^3}.$$

We verify: $|\tilde{g}'''(t)| = \frac{2|d|^3}{(c-dt)^3}$ and $2\tilde{g}''(t)^{3/2} = 2\frac{|d|^3}{(c-dt)^3}$. So $|\tilde{g}'''| = 2(\tilde{g}'')^{3/2}$, confirming self-concordance with equality.

Since the sum of self-concordant functions is self-concordant, ϕ is self-concordant. □

Remark 6.3 (Self-concordance for general constraints). When the constraint functions f_i are nonlinear but convex, the logarithmic barrier may not be self-concordant. However, a rescaled version $c \cdot \phi$ is self-concordant for an appropriate constant c depending on the curvature of the f_i. For convex quadratic constraints, $c = 1$ suffices. The general theory handles this via the *self-concordance parameter* ϑ, which equals m for the standard logarithmic barrier with m linear constraints.

Theorem 6.4 (Newton's method on self-concordant functions)**.** *For a self-concordant function g, Newton's method with appropriate step size (backtracking line search) achieves*

$$g(x_{k+1}) - g^* \leq \epsilon \tag{19}$$

in at most $C_1(g(x_0) - g^*) + C_2 \log\log(1/\epsilon)$ iterations, where C_1 and C_2 are universal constants independent of the problem dimension n and the number of constraints m.

Proof. The proof, due to Nesterov [2004], proceeds in two phases:

(i) **Damped Newton phase:** While the Newton decrement $\lambda(x) = (\nabla g(x)^\top [\nabla^2 g(x)]^{-1} \nabla g(x))^{1/2}$ is large ($\lambda(x) \geq 1/4$), each damped Newton step reduces g by at least a constant. This phase takes $O(g(x_0) - g^*)$ steps.

(ii) **Quadratically convergent phase:** Once $\lambda(x) < 1/4$, full Newton steps converge quadratically: $\lambda(x_{k+1}) \leq 2\lambda(x_k)^2$. This reaches $\lambda(x) < \epsilon$ in $O(\log\log(1/\epsilon))$ steps.

The key property of self-concordant functions is that Newton's method has *affine-invariant* convergence, meaning the iteration count does not depend on the condition number or dimension. □

Theorem 6.5 (Complexity of the barrier method)**.** *Consider problem (1) with m inequality constraints and affine constraint functions f_i. The barrier method with $\sigma = 1 + 1/\sqrt{m}$ and initial $t^{(0)} = m/f_0(x^{(0)})$ achieves $f_0(x) - p^* \leq \epsilon$ in at most*

$$\boxed{O\left(\sqrt{m}\,\log\frac{m}{\epsilon}\right) \text{ Newton iterations (total).}} \tag{20}$$

Each Newton iteration requires solving a linear system of size $n \times n$, costing $O(n^3)$ operations (or less with sparsity).

Proof. **Step 1: Number of outer iterations.** By Proposition 5.3, the number of outer iterations is

$$K = \left\lceil \frac{\log(m/(t^{(0)}\epsilon))}{\log(1 + 1/\sqrt{m})} \right\rceil. \tag{21}$$

Using $\log(1 + u) \geq u/(1 + u)$ for $u > 0$: $\log(1 + 1/\sqrt{m}) \geq \frac{1/\sqrt{m}}{1+1/\sqrt{m}} \geq \frac{1}{2\sqrt{m}}$ for $m \geq 1$. Thus:

$$K \leq 2\sqrt{m}\,\log\left(\frac{m}{t^{(0)}\epsilon}\right) + 1. \tag{22}$$

Step 2: Newton steps per outer iteration. The crucial insight is that with $\sigma = 1 + 1/\sqrt{m}$, the starting point for each centering step is sufficiently close to the new central path point that Newton's method converges in $O(1)$ iterations (a constant independent of m and n).

More precisely, when t increases from $t^{(\ell)}$ to $\sigma\, t^{(\ell)} = (1 + 1/\sqrt{m})\, t^{(\ell)}$, the central path point shifts by an amount measured by the Newton decrement. For the standard logarithmic barrier (a self-concordant function with *self-concordance parameter* $\vartheta = m$), the shift in the Newton decrement is bounded by $\sqrt{m} \cdot |\sigma - 1| = \sqrt{m} \cdot 1/\sqrt{m} = 1$. When the initial Newton decrement is at most 1, the quadratically convergent phase of Newton's method starts immediately, requiring $O(\log\log(1/\epsilon'))$ steps where ϵ' is the centering accuracy.

For the purpose of the barrier method, centering to moderate accuracy suffices (e.g., Newton decrement below 1/4), which requires $O(1)$ Newton steps per outer iteration.

Step 3: Total count. Total Newton iterations $= K \times O(1) = O(\sqrt{m}\log(m/(t^{(0)}\epsilon))) = O(\sqrt{m}\log(m/\epsilon))$ (absorbing $t^{(0)}$ into the constant). □

Barrier Method Complexity

The barrier method solves a convex program with m inequality constraints to ϵ-accuracy in $O(\sqrt{m}\log(m/\epsilon))$ Newton iterations. Key features:

- The dependence on ϵ is $\log(1/\epsilon)$ (polynomial-time), not $1/\epsilon$ as in first-order methods.
- The dependence on m is $\sqrt{m}$, which is optimal for the class of self-concordant barriers.
- Each Newton step costs $O(n^3)$ in general, or $O(n^2 m)$ when the Hessian has special structure.

Remark 6.6 (Comparison with first-order methods).

Method	**Cost per iteration**	**Iterations for ϵ**
Gradient descent	$O(n)$	$O(1/\epsilon)$
Nesterov accel.	$O(n)$	$O(1/\sqrt{\epsilon})$
Barrier method	$O(n^3)$ per Newton step	$O(\sqrt{m}\log(1/\epsilon))$

For small to moderate n and m, interior-point methods dominate due to the logarithmic dependence on $1/\epsilon$. For very large n (as in deep learning), first-order methods are preferred due to the $O(n)$ per-iteration cost.

7 Newton's Method for the Centering Step

Each centering step requires (approximately) solving the unconstrained problem $\min_x\ t\, f_0(x) + \phi(x)$ (we drop equality constraints for clarity). Newton's method is the algorithm of choice due to its quadratic convergence near the optimum.

Definition 7.1 (Newton's method)**.** Given a twice differentiable convex function $g : \mathbb{R}^n \to \mathbb{R}$, Newton's method generates iterates via

$$x_{k+1} = x_k - [\nabla^2 g(x_k)]^{-1} \nabla g(x_k). \tag{23}$$

The *Newton step* $\Delta x_k = -[\nabla^2 g(x_k)]^{-1}\nabla g(x_k)$ is the minimizer of the local quadratic model $g(x_k) + \nabla g(x_k)^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top \nabla^2 g(x_k)(x - x_k)$.

Definition 7.2 (Newton decrement)**.** The *Newton decrement* of g at x is

$$\lambda(x) = \big(\nabla g(x)^\top [\nabla^2 g(x)]^{-1} \nabla g(x)\big)^{1/2} = \|\Delta x\|_{\nabla^2 g(x)}, \tag{24}$$

where $\|v\|_H = (v^\top H v)^{1/2}$ is the local norm induced by the Hessian. The Newton decrement is affine-invariant: it does not change under invertible linear transformations of the variable x.

Remark 7.3 (Newton's method for the barrier). For the centering problem $g(x) = t\, f_0(x) + \phi(x)$, the Newton step requires:

1. Computing the gradient: $\nabla g(x) = t\, \nabla f_0(x) + \nabla \phi(x)$.
2. Computing the Hessian: $\nabla^2 g(x) = t\, \nabla^2 f_0(x) + \nabla^2 \phi(x)$.
3. Solving the linear system $\nabla^2 g(x)\, \Delta x = -\nabla g(x)$.

For linear programs (f_0 linear, f_i linear), the Hessian $\nabla^2 g(x) = \nabla^2 \phi(x)$ has a specific sparse structure that can be exploited for efficient computation.

Proposition 7.4 (Quadratic convergence of Newton's method)**.** *If g is self-concordant and the Newton decrement satisfies $\lambda(x_k) < 1/4$, then the full Newton step (no line search) gives*

$$\lambda(x_{k+1}) \le 2\lambda(x_k)^2. \tag{25}$$

Starting from $\lambda(x_0) < 1/4$, after j Newton steps: $\lambda(x_j) \le 2^{-2^j+1} \cdot \lambda(x_0)$.

Proof. This is a consequence of the self-concordance inequality, which provides affine-invariant control of the third derivative. The detailed proof can be found in Nesterov [2004, Theorem 4.1.14] or Wright [1997, Chapter 4]. □

8 Connection to the MDP LP Formulation

Interior-Point Methods for MDPs

The LP formulation of MDPs. A finite MDP with state space S, action space A, transition probabilities $P(s'|s,a)$, rewards $r(s,a)$, and discount factor $\gamma \in (0,1)$ can be solved via the linear program:

$$\begin{aligned} &\min_{v \in \mathbb{R}^{|S|}} \; c^\top v = \sum_{s \in S} \alpha(s)\, v(s) \\ &\text{s.t. } v(s) \ge r(s,a) + \gamma \sum_{s' \in S} P(s'|s,a)\, v(s'), \quad \forall\, s \in S,\ a \in A, \end{aligned} \tag{26}$$

where α is an initial state distribution (any strictly positive distribution works). The optimal value function v^* is the unique solution. This LP has $|S|$ variables and $|S| \cdot |A|$ inequality constraints.

Applying interior-point methods. The barrier method solves (26) in $O(\sqrt{|S| \cdot |A|}\, \log(|S| \cdot |A|/\epsilon))$ Newton iterations. Each Newton step involves a linear system of size $|S| \times |S|$, costing $O(|S|^3)$ in the dense case. For large MDPs, this is impractical (e.g., $|S| = 10^6$), but for moderate-sized problems it provides an exact solution with guaranteed accuracy.

Dual LP and occupancy measures. The dual of (26) is

$$\begin{aligned} &\max_{d \ge 0} \sum_{s,a} d(s,a)\, r(s,a) \\ &\text{s.t. } \sum_a d(s,a) = \alpha(s) + \gamma \sum_{s',a} d(s',a)\, P(s|s',a), \quad \forall\, s \in S, \end{aligned} \tag{27}$$

where $d(s,a)$ is the discounted occupancy measure. The optimal policy is recovered from d^* via $\pi^*(a|s) = d^*(s,a) / \sum_{a'} d^*(s,a')$.

Constrained MDPs and duality gaps. In constrained MDPs, one adds constraints of the form $\sum_{s,a} d(s,a)\, c(s,a) \le \alpha$ to (27). The resulting LP can be solved by interior-point methods. The duality gap m/t from Theorem 4.5 provides a certificate of near-optimality at each step, which is valuable for safe RL applications where constraint satisfaction must be verified.

9 Exercises

Exercise 9.1 (Barrier for box constraints)**.** Consider the box-constrained problem $\min_x f_0(x)$ subject to $\ell_i \le x_i \le u_i$, $i = 1, \dots, n$, where $\ell_i < u_i$.

(a) Write the logarithmic barrier $\phi(x)$ explicitly.

(b) Compute $\nabla\phi(x)$ and $\nabla^2\phi(x)$.

(c) Show that the Hessian $\nabla^2\phi(x)$ is diagonal.

(d) For $f_0(x) = c^\top x$ (linear objective), find the central path $x^*(t)$ by solving the stationarity condition.

Exercise 9.2 (Central path for LP)**.** For the LP min $c^\top x$ subject to $x \ge 0$ (i.e., $f_i(x) = -x_i$ for $i = 1, \dots, n$):

(a) Write the barrier function $\phi(x) = -\sum_{i=1}^n \log(x_i)$.

(b) Show that the central path satisfies $x_i^*(t) = 1/(tc_i)$ when $c_i > 0$ for all i.

(c) Verify that $c^\top x^*(t) - p^* = n/t$ where $p^* = 0$ (the infimum as $x \to 0^+$).

(d) Compute the dual variables $\lambda_i^*(t)$ and verify that $\lambda_i^*(t) \cdot x_i^*(t) = 1/t$ for all i (approximate complementary slackness).

Exercise 9.3 (Self-concordance verification)**.** Verify self-concordance of $g(x) = -\log(x)$ for $x > 0$:

(a) Compute $g'(x)$, $g''(x)$, $g'''(x)$.

(b) Verify $|g'''(x)| = 2\, g''(x)^{3/2}$.

(c) Show that $g(x) = -\log(b - a^\top x)$ for $a \in \mathbb{R}^n$, $b \in \mathbb{R}$ is self-concordant on $\{x : a^\top x < b\}$. Verify the self-concordance condition along any direction v.

Exercise 9.4 (Newton's method on a quadratic)**.** For $g(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$:

(a) Show that Newton's method converges in one step: $x_1 = Q^{-1}b = x^*$.

(b) Compute the Newton decrement $\lambda(x_0)$ and verify that $g(x_0) - g^* = \frac{1}{2}\lambda(x_0)^2$.

(c) Explain why the "one-step" convergence does not contradict the $O(\log\log(1/\epsilon))$ complexity bound.

Exercise 9.5 (Barrier method complexity)**.** A convex program has $m = 100$ inequality constraints.

(a) With $\sigma = 2$, how many outer iterations are needed to go from $t^{(0)} = 1$ to $t \ge m/\epsilon = 10^8$ (i.e., $\epsilon = 10^{-6}$)?

(b) With $\sigma = 1 + 1/\sqrt{m}$, how many outer iterations?

(c) Explain why the optimal $\sigma = 1 + 1/\sqrt{m}$ gives fewer *total* Newton steps despite more outer iterations.

Exercise 9.6 (Duality gap in constrained MDP)**.** Consider a constrained MDP with one safety constraint: $\max_\pi J(\pi)$ subject to $C(\pi) \le \alpha$, where J is the expected return and C is the expected cost under policy π.

(a) Write the Lagrangian $L(\pi, \lambda)$ for this problem (with $\lambda \ge 0$ the dual variable for the cost constraint).

(b) Show that the dual function $g(\lambda) = \max_\pi L(\pi, \lambda)$ can be computed by solving an unconstrained MDP with modified reward $r(s, a) - \lambda c(s, a)$.

(c) In the occupancy-measure LP formulation, the constrained MDP becomes an LP with $|S| \cdot |A| + 1$ constraints ($|S| \cdot |A|$ non-negativity constraints plus the cost constraint). What is the duality gap m/t along the central path, and how does it relate to the constraint violation?

(d) Under what condition does strong duality hold? (Relate to Slater's condition in the occupancy-measure space.)

Exercise 9.7 (Barrier method for SDP)**.** For a semidefinite program (SDP) min $\operatorname{tr}(CX)$ subject to $\operatorname{tr}(A_i X) = b_i$, $X \succeq 0$ (where X is an $n \times n$ symmetric matrix variable):

(a) The natural barrier for $X \succeq 0$ is $\phi(X) = -\log\det(X)$. Verify that this is a valid barrier (convex on the cone of positive definite matrices, going to $+\infty$ as X approaches the boundary).

(b) Show that $\nabla_X \phi(X) = -X^{-1}$.

(c) The self-concordance parameter of this barrier is $\vartheta = n$. What is the complexity of the barrier method for this SDP?

Exercise 9.8 (Interior-point vs. policy iteration)**.** For a finite MDP with $|S|$ states and $|A|$ actions:

(a) The LP formulation (26) has $m = |S| \cdot |A|$ constraints. What is the barrier method complexity in terms of $|S|$ and $|A|$?

(b) Policy iteration converges in at most $|A|^{|S|}/(1-\gamma)$ iterations (worst case), but often converges in very few iterations in practice. Compare with the interior-point complexity.

(c) For what regime of $|S|$, $|A|$, and γ would interior-point methods be preferred over policy iteration?

References

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